HW4

14.

$H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 < \sigma_2^2$

$df_1 = 29$

$df_2 = 29$

$F = \frac{s_2^2}{s_1^2}$

$= \frac{28.2^2}{15.6^2}$

$= 3.268$

$F_{29,29,0.005} = 2.66$

p-value $= P(F_{29,29} \geq 3.268) < 0.005$

$\therefore$ $F = 3.268 > 2.66$ & $P(F \geq 3.268) < 0.005$ so reject H_0

16.

$H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 \neq \sigma_2^2$

$df_1 = 14$

$df_2 = 14$

$F_{14,14,0.10} = 2.01$

$F = \frac{s_1^2}{s_2^2}$

$= \frac{114^2}{109^2}$

$= 1.225$

p-value $= 2P(F_{14,14} \geq 1.225)$

$\approx 2(0.1)$

> 0.2

$\therefore$ $F = 1.225 < 2.01$, $P(F \geq 1.225) > 0.1$

p-value $= 2 \cdot P(F \geq 1.225) > 0.2$

$\therefore$ no reject H_0

22) a) $H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 \neq \sigma_2^2$

$F = \frac{s_2^2}{s_1^2}$

$= \frac{2.96}{1.54}$

$= 1.922$

$df_1 = df_2 = 14$

$F_{14,14,0.1} = 2.01$

$F = 1.922 < 2.01$

$P(F \geq 1.922) > 0.1$

p-value $= 2 \cdot P(F \geq 1.922) > 0.2$

$\therefore$ Non-reject H_0

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22) (Continue)

b) 99% CI: $\left[\frac{s_1^2}{s_2^2} \frac{1}{F_{14,14,0.005}} , \frac{s_1^2}{s_2^2} F_{14,14,0.005} \right]$

$= \left(\frac{1.54}{2.96} \frac{1}{4.25} , \frac{1.54}{2.96} 4.25 \right]$

$= [0.122, 2.211]$

$\therefore$ because 99% CI include 1, so no reject H_0

11.

$\bar{d} = \frac{-6}{8}$

$= -0.75$

$\mu_d = \mu_1 - \mu_2$

$H_0 : \mu_d = 0$

$H_a : \mu_d \neq 0$

$s_d^2 = \frac{1}{n-1} \left[\sum_{i=1}^{n} d_i^2 - n\bar{d}^2 \right]$

$= \frac{1}{7} (13.5)$

$= 1.9286$

$s_d = \sqrt{1.9286}$

$= 1.389$

$t = \frac{\bar{d} - \mu_0}{s_d / \sqrt{n}}$

$= \frac{-0.75}{1.389/\sqrt{8}}$

$= -1.53$

$df = 7$

$t_{df, \frac{\alpha}{2}} = t_{7, 0.025}$

$= 2.3646$

$RR = \{ t > 2.3646 \}$

$\because |-1.53| = 1.53 < 2.3646$

so non-reject H_0

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12. $\bar{d} = \frac{-6}{8}$

$= -0.75$

$H_0 : \mu_d = 0$

$H_a : \mu_d \neq 0$

$$s_d^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n}(d_i^2 - n\bar{d}^2)\right]$$

$$= \frac{1}{7}\left[10 - (8 \times (-0.75)^2\right]$$

$$= 0.7857$$

$$s_d = \sqrt{0.7857}$$

$$= 0.886$$

$$t = \frac{-0.75}{0.886/\sqrt{8}}$$

$$= -2.394$$

$df = 7$

$|t_{7,0.025}| = 2.3646$

$RR = \{T > 2.3646\}$

$\because |-2.394| = 2.394 > 2.3646$

$\therefore$ 기각 H_0.

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